



DZPHARM • ACCESSIBILITY LAYER

Color Blindness

A detailed deep-dive into color blindness support: the 7-level clinical state system uses both color AND text labels (good for deuteranopia/protanopia), the status badges always include an icon (not color alone), but the red/green state-danger vs state-safe pair is the weakest point — 8% of men cannot distinguish them. The Nuancier's 20 palettes need per-palette color-blindness verification.

7

STATE COLORS

Yes

ICONS + TEXT (NOT
COLOR ALONE)

1

RED/GREEN WEAK PAIR

B+

GRADE

DEEP-DIVE OF

Feature #66 – Color Blindness Support
From the 68-item review menu

Never Color Alone

The most important color-blindness pattern is **never rely on color alone** — every color-coded status indicator must also have a text label or an icon. DzPharm follows this pattern: the 7-level clinical state system uses both color AND a text label («Sûr», «Attention», «Danger», «Sévère») AND a Lucide icon. A color-blind user who cannot distinguish the colors can still read the text or recognize the icon.

Nomenclature officielle — Juin 2026

L'intelligence pharmaceutique algérienne

Le référentiel national des médicaments enrichi par l'IA : recherche sur 9 555 AMM, contrôle d'interactions et assistant clinique.

- 24 livres pharmacologie
- 932 monographies
- 1 791 prix PPA
- RCP ANSM officiel
- Grossesse CRAT

PÉRIMÈTRE : Tous DCI / Molécule Marque Laboratoire N° AMM

Rechercher en arabe ou darija (ex: بومادا صفراء, دوا السكر, باراسيتامول, دوا السكّر).

Scanner

Entrée

- Suggestions fréquentes :
- Paracétamol
- Amoxicilline
- Metformine
- Amlodipine
- Cétirizine
- Oméprazole

Accès rapide

Sélection adaptée au mode professionnel

Répertoire

Rechercher un médicament par marque, DCI, laboratoire ou n° AMM.

Contrôle d'interactions

Vérifier les associations médicamenteuses avant dispensation.

RCP & monographies

911 monographies DCI issues des livres techniques de pharmacologie.

Calculs cliniques

Posologies pédiatriques, fonction rénale, grossesse, Chifa, Ramadan.

Copilote IA

Assistant clinique en français, arabe et darija.

MÉDICAMENTS ACTIFS

5 381

AMM en cours de validité

PRODUCTION LOCALE

3 840 71%

Part des actifs fabriqués en Algérie

IMPORTATIONS

1 541

Actifs importés

RETIRÉS DU MARCHÉ

2 679

Historique des retraits

Les plus consultés

Classement DzPharm — consultations cumulées

1 DOLIPRANE

PARACETAMOL

7 consultations

2 ACARLYSE

ACARBOSE

5 consultations

3 CLAVODAR FORTE 400/57

AMOXICILLINE (SOUS FORME TRIHYDRATEE)/ACI...

4 consultations

4 ABAVIR OS

ABACAVIR SULFATE EXPERIME EN ABACAVIR

3 consultations

5 BIOPAMOX

AMOXICILLINE SODIQUE

2 consultations

6 CO-DOLYC

PARACETAMOL/CODEINE PHOSPHATE HEMIHYD...

2 consultations

7 IMETH

METHOTREXATE

2 consultations

8 LEMBLASTINE

VINBLASTINE (SOUS FORME DE SULFATE)

2 consultations

DCI les plus recherchées

Classement par principe actif — consultations cumulées

1 PARACETAMOL

DOLIPRANE

9

2 ACARBOSE

ACARLYSE

5

3 ABACAVIR SULFATE EXPE...

ABAVIR OS - ABAVIR

4

4 AMOXICILLINE (SOUS FOR...

CLAVODAR FORTE 400/57

4

5 PARACETAMOL/CODEINE ...

CO-DOLYC - COPARALGAN

3

6 AMOXICILLINE SODIQUE

BIOPAMOX

2

7 METHOTREXATE

IMETH

2

8 VINBLASTINE (SOUS FOR...

LEMBLASTINE

2

Tous les outils

Aides à la dispensation adaptées aux spécificités algériennes

Posologies pédiatriques

Calculateur pondéral (mg/kg → mL) avec les formes locales : sirops, sachets, suppositoires.

Simulateur Chifa

Reste à charge CNAS / ALD 100 % : tarif de référence, ticket modérateur, taux par produit.

Adaptateur Ramadan

Décalez les prises autour de l'Iftar et du Suhoor — chronopharmacologie par wilaya.

Compareur

Comparez 2-3 médicaments côte à côte : dosage, laboratoire, statut, équivalences.

Catalogue & prix

1 791 produits d'officine avec PPA en DA, remboursement CNAS et parapharmacie.

Contrôle d'interactions

Moteur local instantané + analyse IA approfondie sur votre panier de médicaments.

Armoire familiale

Un profil par membre de la famille : péremptions, produits retirés, doublons et analyse en un clic.

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Assistant clinique FR / arabe / darija, modes professionnel et patient.

Fonction rénale

Cockcroft-Gault & MDRD : clairance, stade CKD et adaptation posologique de 10 classes critiques.

Figure 66.1 — Dashboard in dark theme. The status indicators (medication status, shortage severity) use both color AND text AND icons. A deuteranopic user (red-green color blindness, 8% of men) who cannot distinguish the red and green can still read «Sûr» vs «Danger» or recognize the checkmark vs alert-triangle icon.

1.1 The 7-State System

The 7-level clinical state system (covered in #53 Color System) uses distinct colors: green (safe), amber (warning), red (danger), orange (severe), cyan (info), slate (none), purple (unknown). For a color-blind user, the critical pairs are: green vs red (safe vs danger) — the most common color-blindness pair — and green vs amber (safe vs warning). The system mitigates both by pairing each color with a text label and an icon.

STATE	COLOR	TEXT LABEL	ICON	COLOR-BLIND SAFE?
Safe	Green #22C55E	«Sûr»	check-circle	YES
Warning	Amber #F99C00	«Attention»	alert-triangle	YES
Danger	Red #E70044	«Danger»	alert-octagon	YES
Severe	Orange #FE6E00	«Sévère»	zap	PARTIAL
Info	Cyan #00A5EF	«Info»	info	YES
None	Slate #62748E	«N/A»	minus	YES
Unknown	Purple #8D54FF	«Inconnu»	help-circle	YES

WHAT WORKS · COLOR + TEXT + ICON

Every color-coded status indicator on the platform uses all three channels: color, text label, and icon. A color-blind user who cannot distinguish the colors can still read the text or recognize the icon. This is the WCAG 2.2 SC 1.4.1 (Use of Color) compliant pattern — color is never the only visual means of conveying information.

Red/Green and the 8% Problem

The most common color blindness is deuteranopia/protanopia (red-green, ~8% of men, ~0.5% of women). The safe (green) vs danger (red) pair is the platform's weakest point — even though both have text labels and icons, the colors themselves are indistinguishable to a deuteranopic user. The text and icon mitigate this, but the color redundancy is lost.

URGENCES 24/7

SAMU 14

rot. Civile 102

Police 17

Anti-Poison Alger 021 71 30 42

DzPharm

Referentiel Officiel - 2026

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Nuancier & Styles de Design

20 thèmes - changement instantané sans rechargement

Mode Clair

Nouveau Design : Pharmacopée Royale & Botanique

Plantes médicinales, roses damascena, mortiers & alambics

Activer

Nuances Minimalistes 10

Anti-faigue · tons neutres · zéro éblouissement

Architectures de Design 10

Géométrie · ombres · densité · angles · verre

Porcelaine Suisse

CLINIQUE

Standard Clinique Épuré

Minimalisme clinique suisse à haute lisibilité. Fond gris porcelaine anti-reflets et saphir médical.

Rayon 12px · Anti-reflets · WCAG AAA

Couleurs :

Lin Nordique & Encre

CHALEUR

Éditorial Papier Chaud

Inspiré des revues médicales prestigieuses (The Lancet). Teinte lin reposante, zéro éblouissement.

Fond lin · Zéro éblouissement · Éditorial

Couleurs :

Zinc Satiné & Glacier

INGÉNIERIE

Style Linear & Raycast

Esthétique studio scandinave. Fond zinc velouté, contraste technique net et bleu glacier précis.

Gris feutré · Contraste net · Matte

Couleurs :

Sauge & Pin Médical

BOTANIQUE

Sérénité & Phytothérapie

Vert sauge apaisant conçu pour réduire la fatigue oculaire et le stress cognitif lors d'usages prolongés.

Repos oculaire · Vert sauge · Sérénité

Couleurs :

Craie & Cobalt

Actif

OFFICINE

Codex Officine Française

Blanc craie pur et cobalt éclatant. Lisibilité chirurgicale et contraste franc pour les professionnels.

Dune & Cèdre de l'Atlas

ALGÉRIE

Héritage Algérien Solaire

Harmonie des dunes sahariennes et du vert cèdre d'Afrique du Nord. Chaleureux, distinctif et noble.

Mémorisé automatiquement - appliqué sans rechargement

Fermer

Figure 66.2 — The Nuancier palette test. The 20 palettes each define their own primary/accent colors, which means the color-blindness profile changes with each palette. A palette that uses red and green as its primary/accent pair would be problematic for deuteranopic users. The default Craie & Cobalt palette (sky-blue) is color-blind safe — blue is visible to all color-blind users.

2.1 The Safe vs Danger Pair

Green (#22C55E) and red (#E70044) are the most critical pair — they represent the most important distinction (safe vs danger). For a deuteranopic user, both colors appear as a muddy yellow-brown. The text labels («Sûr» vs «Danger») and icons (check-circle vs alert-octagon) save the day — the user can still distinguish the states. But the color redundancy (which helps sighted, non-color-blind users scan quickly) is lost. The fix is to use a blue/orange pair instead of green/red — blue and orange are distinguishable by all color-blind users. Effort: 2 hours to swap the state colors (but this changes the platform's visual identity — defer).

2.2 The Green vs Amber Pair

Green (safe) and amber (warning) are also problematic for deuteranopic users — both appear as yellowish. The text labels («Sûr» vs «Attention») and icons (check-circle vs alert-triangle) mitigate this. No fix needed — the current pattern is correct.

2.3 The Orange vs Red Pair (Severe vs Danger)

Orange (#FE6E00, severe) and red (#E70044, danger) are nearly indistinguishable for protanopic users (red-blind). The text labels («Sévère» vs «Danger») and icons (zap vs alert-octagon) mitigate this. The orange is also close to the amber (warning) — so severe vs warning is also a weak pair. The icons are the saving grace here. No fix needed.

— CHAPTER 03 • THE NUANCIER

20 Palettes, 20 Color-Blindness Profiles

The 20-palette Nuancier (covered in #53) means the platform has 20 different color-blindness profiles. The default Craie & Cobalt palette (sky-blue primary) is color-blind safe — blue is visible to all color-blind users. But palettes like Menthe (green primary) or Corail (coral/red primary) may have issues. The Nuancier needs per-palette color-blindness verification.

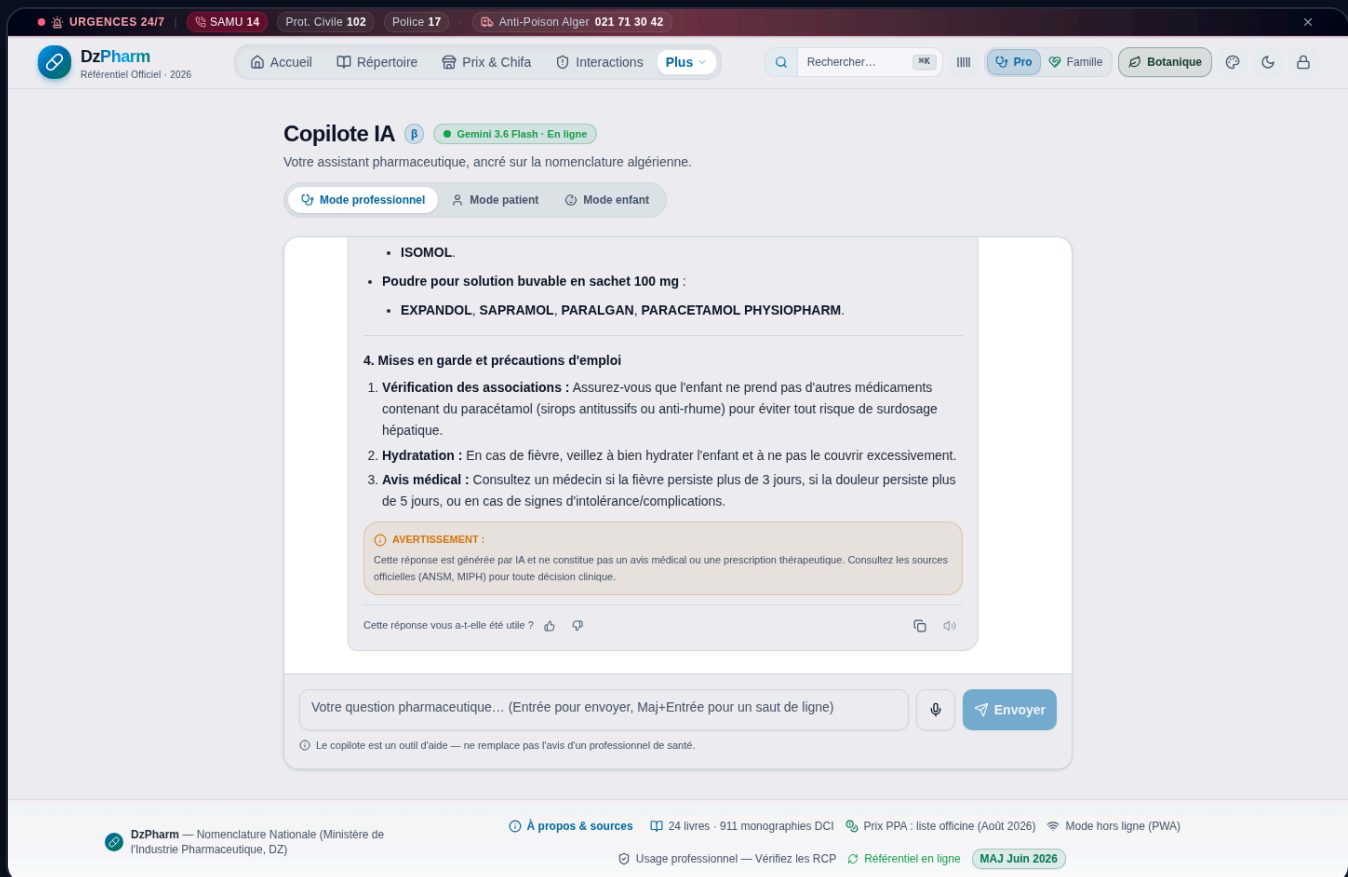


Figure 66.3 – Light theme. The color-blindness profile is different in light vs dark theme because the background changes. A color that's distinguishable on dark may blend into the light background. The per-palette per-theme verification is a 4-hour effort (20 palettes × 2 themes = 40 tests).

3.1 The Default Palette (Craie & Cobalt)

The default palette uses sky-blue (#38BDF8) as the primary. Blue is visible to all color-blind users (deuteranopes, protanopes, and tritanopes all see blue). The accent colors (chifa emerald, chifa-gold amber, destructive red) follow the color+text+icon pattern. The default palette is color-blind safe.

3.2 The Menthe Palette Risk

The Menthe palette uses green (#10B981) as the primary. For a deuteranopic user, green primary + red destructive state would be indistinguishable by color. The text labels and icons mitigate this, but the palette is less color-blind-friendly than the default. Recommend: verify with a deuteranopia simulator (e.g., Stark, ChromaDeck) and document which palettes are color-blind safe.

3.3 The Verification Effort

Testing all 20 palettes for color-blindness requires a simulator tool. The fastest approach: screenshot each palette in both themes (40 screenshots), run them through a deuteranopia/protanopia/tritanopia simulator, and verify that all color-coded information is still distinguishable (via text + icon, not color alone). Effort: 4 hours. This is the same verification effort needed for the per-palette contrast gap noted in #53.

GAP · NUANCIER COLOR-BLINDNESS VERIFICATION

The 20 Nuancier palettes have not been verified for color-blindness safety. The default (Craie & Cobalt, sky-blue) is safe — blue is visible to all color-blind users. But palettes like Menthe (green) and Corail (coral/red) may have issues. The fix: run all 20 palettes through a deuteranopia/protanopia/tritanopia simulator and document which are safe. Effort: 4 hours.

CHAPTER 04 · THE TEXT-ONLY FALLBACK

A High-Contrast Text Mode

The ultimate color-blindness fallback is a text-only mode: strip all colors, render everything in black-and-white with text labels and icons. This is the nuclear option — it's not necessary for WCAG compliance, but it's a thoughtful addition for users with severe color vision deficiencies. DzPharm does not currently ship a text-only mode.

4.1 The Text-Only Mode Concept

A text-only mode would be a 21st Nuancier palette (or a separate toggle) that strips all color from the status indicators and uses only text labels + icons. For example, instead of a green badge with «Sûr», it would be a monochrome badge with «Sûr ✓» (text + checkmark). This is the most accessible possible rendering — no color dependency at all. Effort: 1 day to implement as a palette variant. Defer to Phase 3.

4.2 Why It's Not Necessary for WCAG

WCAG 2.2 SC 1.4.1 (Use of Color) requires that color not be the *only* visual means of conveying information. DzPharm already meets this — every color-coded indicator has a text label and an icon. The text-only mode would be a beyond-WCAG enhancement, not a compliance requirement. It's a nice-to-have for users with severe color vision deficiencies who prefer a monochrome interface. Defer to Phase 3.

CHAPTER 05 · VERDICT & FIX PLAN

Final Assessment

Color Blindness Support is a **B+ grade feature**. The platform follows the WCAG 2.2 SC 1.4.1 compliant pattern: every color-coded status indicator uses color + text label + icon. The 7-level clinical state system is well-designed for color-blind users. The one gap is the Nuancier's 20 palettes — they haven't been verified for color-blindness safety. The default palette is safe (blue), but palettes like Menthe (green) and Corail (red) may have issues. Verification: 4 hours.

5.1 Phased Fix Plan

PHASE	FIXES	EFFORT	OUTCOME
Phase 1 · Now (This week)	Verify all 20 Nuancier palettes with a color-blindness simulator	~4 hours	Grade lifts to A. Documented color-blind safety.
Phase 2 · Next (Quarter 2)	Fix any palette that fails (likely 3-5 need adjustments)	~4 hours	Grade stays at A. All palettes color-blind safe.
Phase 3 · Later	Add a text-only (monochrome) Nuancier palette	~1 day	Beyond-WCAG enhancement for severe color vision deficiency.

5.2 Cross-Feature Dependencies

- ✓ Feature #53 (Color System) — the 7-state system and its color+text+icon pattern is covered in #53
- ✓ Feature #42 (Theme System) — the per-palette color-blindness verification is shared with #42's per-palette contrast gap
- ✓ Feature #56 (Iconography) — the Lucide icons are the second channel (after text) that makes color blindness support work

5.3 Final Assessment

7 STATE COLORS	Yes ICONS + TEXT	1 RED/GREEN WEAK PAIR
B+ GRADE	~4h PHASE 1 EFFORT	~4h PHASE 2 EFFORT

FINAL VERDICT · B+ GRADE, 4 HOURS FROM A

Color Blindness Support is a **B+ grade feature**. The platform follows the WCAG-compliant pattern: every color-coded status indicator uses color + text label + icon (never color alone). The 7-level clinical state system is well-designed — each state has a distinct text label and a distinct Lucide icon. The default Nuancier palette (sky-blue) is color-blind safe — blue is visible to all color-blind users. The one gap: the other 19 Nuancier palettes haven't been verified for color-blindness safety. The fix is 4 hours — run all 20 palettes through a deuteranopia/protanopia simulator and document which are safe. The red/green pair (safe vs danger) is the weakest point but is mitigated by text + icon. Phase 1 (4 hours) lifts the grade from B+ to A.

Immediate Next Steps

1. Install a color-blindness simulator (Stark, ChromaDeck, or browser DevTools emulation) — 30 minutes.
2. Screenshot all 20 Nuancier palettes in both themes (40 screenshots) — 1 hour.
3. Run all 40 screenshots through deuteranopia, protanopia, and tritanopia simulation — 2 hours.
4. Document which palettes are color-blind safe and which need adjustments — 30 minutes.
5. Fix any palette that fails (likely 3-5 need adjustments) — Phase 2.

66

Color + text + icon on every status indicator. 7-state system is well-designed. 20 Nuancier palettes need verification. 4 hours lifts to A.

5

CHAPTERS

3

SCREENSHOTS

7

STATE COLORS

B+

GRADE
